Supplementary Materials for "A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology"

This supplementary materials includes proofs of technical results and additional numerical examples supporting conclusions from Section 4 the main manuscript.

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1 Proofs

1.1 Proof of Theroem 2.2

Proof. Fix a population i and suppress the index i. First, since F_X is continuous, we can express well-defined order statistics as $X_{(1)} < \cdots < X_{(N)}$. By rank-based selection, the observed system consists of indices j such that $r_j \in \mathcal{R}$, yielding observed pairs $(Y_{(j)}, X_{(r_j)})$, for $j = 1, \ldots, n$. By assumption, for each observed rank j there exists a strictly increasing measurable function ψ_j such that

$$Y_{(j)} = \psi_j(X_{(r_j)}) \quad \text{a.s.}$$

We now identify the distribution function $F_{Y_{(j)}}$ for fixed j. Since ψ_j is strictly increasing, it admits a measurable inverse on its range. For any $y \in \mathbb{R}$,

$$\mathbb{P}(Y_{(j)} \leq y) = \mathbb{P}(\psi_j(X_{(r_j)}) \leq y) = \mathbb{P}(X_{(r_j)} \leq \psi_j^{-1}(y)).$$

Hence, $F_{Y_{(j)}}(y) = F_{X_{(r_j)}}(\psi_j^{-1}(y))$, which implies that

$$F_{Y_{(j)}}(\psi_j(x)) = F_{X_{(r_j)}}(x) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

We now show that $Y_{(j)}$ is recovered from the Full House Modeling construction. Since F_X is continuous, the distribution of the order statistic $X_{(r_j)}$ is also continuous and strictly increasing on the interior of its support, hence admits a well-defined generalized inverse. Likewise, $F_{Y_{(j)}}$ is continuous and strictly increasing. Thus both admit well-defined inverses. Let $u = F_{X_{(r_j)}}(x)$. Then $F_{Y_{(j)}}(\psi_j(x)) = u$, so that

$$\psi_j(x) = F_{Y_{(j)}}^{-1}(u) = F_{Y_{(j)}}^{-1}\big(F_{X_{(r_j)}}(x)\big)$$

for $F_{X_{(r_j)}}$-almost every x. Therefore,

$$Y_{(j)} = F^{-1}_{Y_{(j)}} \circ F_{X_{(r_j)}}(X_{(r_j)}) \quad \text{a.s.} \tag{1}$$

We now verify the uniqueness of (1). Let $U = F_{X_{(r_j)}}(X_{(r_j)}) \sim \text{Unif}(0,1)$ by the probability integral transform, and define

$$g(u) := \psi_j\big(F^{-1}_{X_{(r_j)}}(u)\big).$$

Then g is strictly increasing and $g(U) \sim F_{Y_{(j)}}$. By the fact that if $U \sim \text{Unif}(0,1)$ and g is increasing, then the distribution of $g(U)$ uniquely determines g almost everywhere, implying $g(u) = F^{-1}_{Y_{(j)}}(u)$ a.e. It follows that

$$\psi_j(x) = F^{-1}_{Y_{(j)}}\big(F_{X_{(r_j)}}(x)\big) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

Thus, the strictly increasing coupling is unique up to null sets. The representation holds for each observed rank j, and uniqueness implies that the joint distribution of (X, Y) coincides with that induced by the Full House Model. □

1.2 Proof of Lemma 3.2

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k}))}, f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. A similar technique can be applied to the second equation to obtain the result. □

1.3 Proof of Theorem 3.4

Proof. By Lemma 3.2,

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Under the assumption $F_{Y_1} = F_{Y_2} = F_Y$, the distributions of the order statistics admit Beta representations, which allow us to express the mapping $h_{j,k}$ in terms of regularized incomplete beta functions.

$$\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1}\circ I^{-1}_{r_{1,j},N_1-r_{1,j}+1}\circ I_{j,n_1-j+1}\circ I^{-1}_{k,n_2-k+1}\circ I_{r_{2,k},N_2-r_{2,k}+1}\circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1}\circ I^{-1}_{r_{1,j},N_1-r_{1,j}+1}\circ I_{j,n_1-j+1}\circ I^{-1}_{k,n_2-k+1}\circ I_{r_{2,k},N_2-r_{2,k}+1}\circ F_X(x)} d\, I_{r_{1,j},N_1-r_{1,j}+1}(F_X(y)) \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ d\, I_{r_{2,k},N_2-r_{2,k}+1}(F_X(x)) \\
&= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.
\end{aligned}$$

This representation holds without additional assumptions. The positive-part operator appears because the integrand may change sign. Under condition (8), the integrand is nonnegative, hence

$$\begin{aligned}
&\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt \\
&= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\} dt.
\end{aligned}$$

Thus,

$$
\begin{aligned}
&\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, dI_{k,n_2-k+1}(x) - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, dI_{r_{2,k},N_2-r_{2,k}+1}(x) \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)}\, dx \\
&\qquad - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)}\, dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p (1-x)^{n_1-p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)}\, dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)}\, dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2+n_1-p-k+1)}{B(k, n_2-k+1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}
$$

□

1.4 Proof of Proposition 3.5

Proof. Specification of the selection rule $S_{i,j} = 1 \iff X_{i,j} \geq X_{i,(N_i-n_i+1)}$ implies that $r_{i,j} = N_i + 1 - j$. When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $t \in (0,1)$, that

$$
I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.
$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased. This follows from the monotonicity of the inverse incomplete beta function in its parameters [Askitis, 2021], together with the fact that stochastic dominance is preserved under monotone transformations. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$

and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned} & I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t. \end{aligned}$$

Technical condition (8) holds and the conclusion of Theorem 3.4 applies. □

1.5 Proof of Proposition 4.2

Proof. First, in combination with the Definition 4.1 and the assumption $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, we have

$$F_{Y_1}(x)^{\frac{n_1}{N_1}} \geq F_{Y_2}(x)^{\frac{n_2}{N_2}} \implies x \geq F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right).$$

By applying the Equation 7 from the Lemma 3.2 we have

$$\begin{aligned} & \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ & = \mathrm{P}\left(F^{-1}_{Y_1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ & = \int_{-\infty}^{+\infty} \int_{F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x) \\ & = \int_{-\infty}^{+\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\ & = \int_0^1 F_{Y_1}\left(F^{-1}_{Y_2}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt. \end{aligned}$$

Based on the Definition 4.1, we have

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \int_0^1 F_{Y_1}\left(F^{-1}_{Y_2}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt$$

$$
\begin{aligned}
&\geq \int_0^1 F_{Y_2}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}}\,dt \\
&= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},
\end{aligned}
$$

and

$$
\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}}\,dt \\
&\leq \int_0^1 \left(t^{1/n_2}+\varepsilon\right)^{n_1} - t^{\frac{N_1}{N_2}}\,dt \\
&= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}.
\end{aligned}
$$

□

1.6 Proof of Theorem 4.3

Proof. Equation 7 from Lemma 3.2 yields

$$
g_{n_1,n_2}(x) = F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right) = \Phi^{-1}\left(\Phi\left(\frac{x-\mu_2}{\sigma_2}\right)^{\frac{n_2N_1}{N_2n_1}}\right)\sigma_1 + \mu_1.
$$

Let A and R be as defined, respectively, in (11) and (13). We have,

$$
\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2N_1}{N_2n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y)dF_{Y_2,(n_2)}(x) \\
&= \int_A F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\,dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\,dF_{Y_2,(n_2)}(x) - \int_{A^c} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\,dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\,dF_{Y_2,(n_2)}(x) + R.
\end{aligned}
$$

We continue with the left-hand side of the last equality,

$$\begin{aligned}
\int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_{2,(n_2)}}(x) &= \int_0^1 \left[F_{Y_1}\left(F_{Y_2}^{-1}(t^{1/n_2})\right)^{n_1} - t^{N_1/N_2}\right] dt \\
&= \int_0^1 \left(\Phi\left(\frac{\Phi^{-1}(t^{1/n_2})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}\right)\right)^{n_1} dt \;-\; \frac{N_2}{N_1+N_2} \\
&= \int_{-\infty}^{+\infty} \Phi\left(\frac{\sigma_2 x + \mu_2 - \mu_1}{\sigma_1}\right)^{n_1} \left(n_2 \Phi(x)^{n_2-1}\phi(x)\right) dx \;-\; \frac{N_2}{N_1+N_2} \\
&= n_2 \int_{-\infty}^{+\infty} \mathbb{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \;\middle|\; X = x \right) \phi(x)\, dx \;-\; \frac{N_2}{N_1+N_2} \\
&= n_2\, \mathbb{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \right) \;-\; \frac{N_2}{N_1+N_2} \\
&= n_2\, F_{\mathbf{W}}\left(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)\right) \;-\; \frac{N_2}{N_1+N_2}.
\end{aligned}$$

The distribution function $F_{\mathbf{W}}$ corresponds to random variable $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ where $\mathbf{\Sigma}$ is as defined in (2) and (3). The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0.

The bound on the remainder (17) follows from the argument in Section 4.2. □

1.7 Proof of Theorem 4.4

Proof.

$$\begin{aligned}
\mathbf{F}_{\mathbf{W}}(\mathbf{w}) &= \mathbb{P}\left(\begin{array}{l} Z_1 + \frac{\sigma_2}{\sigma_1} Z_0 \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{n_1} + \frac{\sigma_2}{\sigma_1} Z_0 \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{n_1+1} + Z_0 \le 0, \ldots, Z_{n_1+n_2-1} + Z_0 \le 0 \end{array} \right) \\
&= \mathbb{P}\left(\begin{array}{l} Z_1 \le \frac{\mu_2-\mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \ldots, Z_{n_1} \le \frac{\mu_2-\mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \\ Z_{n_1+1} \le -Z_0, \ldots, Z_{n_1+n_2-1} \le -Z_0 \end{array} \right)
\end{aligned}$$

Define $M_{n_1} = \max\{Z_1, Z_2, \ldots, Z_{n_1}\}$, $M_{n_2-1} = \max\{Z_{n_1+1}, Z_{n_1+2}, \ldots, Z_{n_1+n_2-1}\}$, then the CDF of $\mathbf{W}$ essentially computes the joint probability:

$$P(M_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0 \text{ and } M_{n_2-1} \leq -Z_0)$$

From extreme value theory, as $n_1 \to \infty$, M_{n_1} sharply concentrates around $\sqrt{2 \ln n_1}$ [Embrechts et al., 2013].

Substitute this into the first inequality:

$$\sqrt{2 \ln n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0 \implies Z_0 \leq \frac{\frac{\mu_2 - \mu_1}{\sigma_1} - \sqrt{2 \ln n_1}}{\frac{\sigma_2}{\sigma_1}}$$

As n_1 grows, $\frac{\frac{\mu_2 - \mu_1}{\sigma_1} - \sqrt{2 \ln n_1}}{\frac{\sigma_2}{\sigma_1}}$ becomes a very large negative number. This forces the random variable Z_0 deep into its left tail.

Similarly, as $n_2 \to \infty$, M_{n_2-1} sharply concentrates around $\sqrt{2 \ln(n_2 - 1)}$ [Embrechts et al., 2013].

Substitute this into the second inequality:

$$\sqrt{2 \ln(n_2 - 1)} \leq -Z_0 \implies Z_0 \leq -\sqrt{2 \ln(n_2 - 1)}$$

As n_2 grows, $-\sqrt{2 \ln(n_2 - 1)}$ becomes a very large negative number. This forces the random variable Z_0 deep into its left tail.

Then the CDF of $\mathbf{W}$ essentially computes the probability:

$$\mathbf{P}(Z_0 \leq \frac{\frac{\mu_2 - \mu_1}{\sigma_1} - \sqrt{2 \ln n_1}}{\frac{\sigma_2}{\sigma_1}}, Z_0 \leq -\sqrt{2 \ln(n_2 - 1)}),$$

as $n_1, n_2 \to \infty$.

This probability is dominated entirely by the tail of Z_0. We can use the Mill's ratio [Mills,

1926] and apply it to standard Gaussian tail approximation, $\Phi(-x) \sim \frac{\phi(x)}{x}$ as $x \to \infty$.

Since

$$\lim_{n_1,n_2\to\infty} \frac{\ln n_2}{\ln n_1} > \left(\frac{\sigma_1}{\sigma_2}\right)^2,$$

the tail behavior of Z_0 is dominated by n_2, then

$$F_{\mathbf{W}}(\mathbf{w}) \approx \Phi(-\sqrt{2\ln(n_2-1)}) \sim \frac{\phi(\sqrt{2\ln(n_2-1)})}{\sqrt{2\ln(n_2-1)}} = \frac{1}{\sqrt{4\pi\ln(n_2-1)}}\left(\frac{1}{n_2-1} + o(\frac{1}{n_2})\right)$$

Let $\mathbf{W}^* \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma^*})$ and covariance matrix

$$\mathbf{\Sigma^*} = \begin{bmatrix} \Sigma^*_{11} & \Sigma^*_{12} \\ \Sigma^*_{21} & \Sigma^*_{22} \end{bmatrix}, \tag{2}$$

defined with the following structure

$$\Sigma^*_{11} = \begin{bmatrix} \frac{\sigma_4^2}{\sigma_3^2}+1 & \frac{\sigma_4^2}{\sigma_3^2} & \dots & \frac{\sigma_2^2}{\sigma_3^2} \\ \frac{\sigma_4^2}{\sigma_3^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_4^2}{\sigma_3^2} \\ \frac{\sigma_4^2}{\sigma_3^2} & \dots & \frac{\sigma_4^2}{\sigma_3^2} & \frac{\sigma_4^2}{\sigma_3^2}+1 \end{bmatrix}_{n_1\times n_1} \quad \Sigma^{T*}_{12} = \Sigma^*_{21} = \begin{bmatrix} \frac{\sigma_4}{\sigma_3} & \dots & \dots & \frac{\sigma_4}{\sigma_3} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_4}{\sigma_3} & \dots & \dots & \frac{\sigma_4}{\sigma_3} \end{bmatrix}_{(n_2-1)\times n_1}$$

$$\Sigma^*_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)} . \tag{3}$$

and $\mathbf{w}^* = (\frac{\mu_4-\mu_3}{\sigma_3}, \dots, \frac{\mu_4-\mu_3}{\sigma_3}, 0, \dots, 0)$, where the first n_1 elements of $\mathbf{w}^*$ are $\frac{\mu_4-\mu_3}{\sigma_3}$ and the last n_2-1 elements are 0.

Then the $\mathbf{F}_{\mathbf{W}^*}(\mathbf{w}^*)$ is

$$\mathbf{F}_{\mathbf{W}^*}(\mathbf{w}^*) = \mathbb{P}\left(\begin{array}{l} Z_1 + \frac{\sigma_4}{\sigma_3} Z_0 \leq \frac{\mu_4 - \mu_3}{\sigma_3}, \ldots, Z_{n_1} + \frac{\sigma_4}{\sigma_3} Z_0 \leq \frac{\mu_4 - \mu_3}{\sigma_3}, \\ Z_{n_1+1} + Z_0 \leq 0, \ldots, Z_{n_1+n_2-1} + Z_0 \leq 0 \end{array} \right)$$
$$= \mathbb{P}\left(\begin{array}{l} Z_1 \leq \frac{\mu_4 - \mu_1}{\sigma_3} - \frac{\sigma_4}{\sigma_3} Z_0, \ldots, Z_{n_1} \leq \frac{\mu_4 - \mu_1}{\sigma_3} - \frac{\sigma_4}{\sigma_3} Z_0, \\ Z_{n_1+1} \leq -Z_0, \ldots, Z_{n_1+n_2-1} \leq -Z_0 \end{array} \right)$$

Since the μ_3, μ_4, σ_3, and σ_4 satisfy the same assumptions as μ_1, μ_2, σ_1, and σ_2 in the theorem statement, and applying the same technique for $\mathbf{F}_{\mathbf{W}}(\mathbf{w})$, we have

$$F_{\mathbf{W}^*}(\mathbf{w}^*) \approx \Phi(-\sqrt{2\ln(n_2 - 1)}) \sim \frac{\phi(\sqrt{2\ln(n_2 - 1)})}{\sqrt{2\ln(n_2 - 1)}} = \frac{1}{\sqrt{4\pi \ln(n_2 - 1)}} \left(\frac{1}{n_2 - 1} + o(\frac{1}{n_2}) \right)$$

Consider the difference between paradox probabilities using two sets of different parameters, and

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \mu_3, \mu_4, \sigma_3, \sigma_4 \right) \\ &\quad - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \mu_1, \mu_2, \sigma_1, \sigma_2 \right) \\ &= n_2 (F_{\mathbf{W}^*}(\mathbf{w}^*) - F_{\mathbf{W}}(\mathbf{w})) + (R^* - R) \to 0 \end{aligned}$$

as $n_1, n_2 \to \infty$. where $R^* := R(\mu_3, \mu_4, \sigma_3, \sigma_4, n_1, n_2, N_1, N_2)$ and $R := R(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$ are defined in (13) □

2 Simulations

2.1 Simulation Results for Theorem 4.3

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.3 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (2) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.3 using mvtnorm package [Genz and Bretz, 2009] in R.

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

n_1	n_2	N_1	N_2	μ_1	μ_2	σ_1	σ_2	Simulation Prob	Theoretical Prob
10	40	400	600	-1	1	1.0	3	0.4000	0.4000099
10	20	400	600	0	0	1.0	3	0.3970	0.3970305
10	40	400	300	-1	1	1.0	3	0.5715	0.5714174
10	20	400	600	0	1	1.5	2	0.3645	0.3647783
10	40	400	600	0	1	1.0	3	0.3995	0.3999245
20	40	400	600	0	1	1.0	3	0.3995	0.3999803
10	40	400	600	-1	0	1.5	3	0.4000	0.3995149
20	40	400	600	0	0	1.5	2	0.2970	0.2962431
20	40	400	600	0	0	1.0	3	0.3750	0.3740938
10	20	400	600	-1	1	1.0	3	0.3990	0.4000059

Table 1: Simulation Results for Normal Example.

References

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Theorem 2.1 (Asymptotic stability under plug-in estimation)**.** *Assume a Full House Model as in Definition 2.1 and suppose that* $Y_i \sim N(\mu_i, \sigma_i^2)$*,* $i = 1, 2$*. Assume that the selection indicators satisfy* $S_{i,(N_i)} = 1$ *for* $i = 1, 2$*. Let the paradox probability be given by Theorem 4.3.*

Let $\hat{\mu}_i$ *and* $\hat{\sigma}_i$ *be estimators satisfying*

$$\hat{\mu}_i - \mu_i = O_P(n_i^{-1/2}), \qquad \hat{\sigma}_i - \sigma_i = O_P(n_i^{-1/2}), \quad i = 1, 2.$$

Suppose that $n_1, n_2 \to \infty$ *with*

$$\frac{\ln n_2}{\ln n_1} > \left(\frac{\sigma_1}{\sigma_2}\right)^2,$$

and that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $\sigma_1 \leq \sigma_2$, *and* $\mu_1 \leq \mu_2$.

Then the plug-in estimator of the paradox probability is consistent, i.e.

$$P(\hat{\theta}) - P(\theta) \xrightarrow{P} 0,$$

where $\theta = (\mu_1, \mu_2, \sigma_1, \sigma_2)$.

2.2 Proof of Theorem 2.1

Proof. By Theorem 4.3, the paradox probability admits the decomposition

$$P(\theta) = n_2 F_{\mathbf{W}}(\mathbf{w}(\theta)) - \frac{N_2}{N_1 + N_2} + R(\theta),$$

where $R(\theta)$ is the remainder term defined in (13). Let $\hat{\theta}$ denote the plug-in estimators. Then

$$P(\hat{\theta}) - P(\theta) = n_2\Big(F_{\mathbf{W}}(\mathbf{w}(\hat{\theta})) - F_{\mathbf{W}}(\mathbf{w}(\theta))\Big) + \big(R(\hat{\theta}) - R(\theta)\big).$$

We analyze the two terms separately.

Step 1: Control of the remainder term.

From (17), under the stated assumptions,

$$|R(\theta)| \leq \left(\frac{1}{2}\right)^{n_2}.$$

Since this bound depends only on (n_1, n_2), it also holds for $R(\hat{\theta})$ for sufficiently large samples.

Therefore,

$$|R(\hat{\theta}) - R(\theta)| \leq 2\left(\frac{1}{2}\right)^{n_2} \longrightarrow 0.$$

Step 2: Asymptotic evaluation via conditioning.

From the representation established in the proof of Theorem 4.3, we have

$$F_{\mathbf{W}}(\mathbf{w}(\theta)) = \mathrm{P}(M_{n_1} \leq T_1(Z_0),\ M_{n_2-1} \leq T_2(Z_0))\,,$$

where

$$T_1(Z_0) = \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \qquad T_2(Z_0) = -Z_0,$$

and $Z_0, Z_1, Z_2, \ldots$ are independent standard normal random variables.

Conditioning on Z_0 and using independence of the maxima, we obtain

$$F_{\mathbf{W}}(\mathbf{w}(\theta)) = \mathbb{E}\Big[\,\mathrm{P}(M_{n_1} \leq T_1(Z_0)) \cdot \mathrm{P}(M_{n_2-1} \leq T_2(Z_0))\Big].$$

Since $\mathrm{P}(M_n \leq t) = \Phi(t)^n$, this becomes

$$F_{\mathbf{W}}(\mathbf{w}(\theta)) = \mathbb{E}\Big[\Phi(T_1(Z_0))^{n_1} \cdot \Phi(-Z_0)^{\,n_2-1}\Big].$$

We analyze this expectation asymptotically. Let $a_{n_2} := \sqrt{2\ln n_2}$. We decompose the expectation according to the value of Z_0.

(i) Upper tail region: $Z_0 \geq -a_{n_2} + C$.

In this region, $\Phi(-Z_0) \leq \Phi(a_{n_2} - C) < 1$, and standard Gaussian tail bounds imply that

$$\Phi(-Z_0)^{n_2-1} \leq \exp(-cn_2^{\delta})$$

for some constants $c, \delta > 0$. Hence the contribution of this region is negligible.

(ii) Lower tail region: $Z_0 \leq -a_{n_2} - C$.

In this region, $\Phi(-Z_0) \to 1$, so $\Phi(-Z_0)^{n_2-1} \to 1$, but

$$\mathrm{P}(Z_0 \le -a_{n_2} - C) \le \Phi(-a_{n_2} - C) = O\left(\frac{1}{n_2} e^{-Ca_{n_2}}\right),$$

which is exponentially small. Thus this region also contributes negligibly.

(iii) Transition region: $Z_0 = -a_{n_2} + O(1)$.

This region dominates the expectation. Write $Z_0 = -a_{n_2} + u$ with $u = O(1)$. Then

$$\Phi(-Z_0)^{n_2-1} = \Phi(a_{n_2} - u)^{n_2-1} \to \exp(-e^u),$$

by classical extreme-value asymptotics for Gaussian maxima.

For the first factor, we have

$$T_1(Z_0) = \frac{\mu_2 - \mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1}(a_{n_2} - u),$$

so that $T_1(Z_0) \to \infty$ uniformly in this region. Using Gaussian tail bounds,

$$1 - \Phi(T_1(Z_0)) \le \exp\left(-\frac{T_1(Z_0)^2}{2}\right),$$

and hence

$$\Phi(T_1(Z_0))^{n_1} = 1 - o(1),$$

provided

$$n_1 \exp\left(-\frac{T_1(Z_0)^2}{2}\right) \to 0.$$

This holds under the condition

$$\frac{\ln n_2}{\ln n_1} > \left(\frac{\sigma_1}{\sigma_2}\right)^2.$$

Combining the three regions, we obtain

$$F_{\mathbf{W}}(\mathbf{w}(\theta)) = \mathbb{E}\big[\Phi(-Z_0)^{n_2-1}\big] + o(1).$$

Then

$$\mathbb{E}\big[\Phi(-Z_0)^{n_2-1}\big] = \mathbb{E}\big[\,\mathrm{P}(M_{n_2-1} \le -Z_0 \mid Z_0)\big] = \mathrm{P}(M_{n_2-1} \le -Z_0).$$

and standard extreme-value theory yields

$$M_{n_2-1} = \sqrt{2\ln n_2} + O_P(1).$$

Therefore,

$$F_{\mathbf{W}}(\mathbf{w}(\theta)) = \Phi(-\sqrt{2\ln n_2})(1 + o(1)).$$

Step 3: Stability under parameter perturbations.

The mapping $\theta \mapsto \mathbf{w}(\theta)$ is smooth, so

$$\|\mathbf{w}(\hat{\theta}) - \mathbf{w}(\theta)\| = O_P(n_1^{-1/2} + n_2^{-1/2}).$$

From Step 2, $F_{\mathbf{W}}(\mathbf{w}(\theta))$ is asymptotically equivalent to a one-dimensional Gaussian tail probability with threshold $x = \sqrt{2\ln n_2}$. Small perturbations of θ induce perturbations $\delta = O_P(n_1^{-1/2} + n_2^{-1/2})$ in this threshold.

Since $n_i^{-1/2} = o(\sqrt{\ln n_2})$, we may apply the expansion

$$\Phi(-x + \delta) - \Phi(-x) = \phi(x)\delta + o(\delta\phi(x)), \quad x \to \infty.$$

Thus,

$$F_{\mathbf{W}}(\mathbf{w}(\hat{\theta})) - F_{\mathbf{W}}(\mathbf{w}(\theta)) = O_P\Big(\phi(\sqrt{2\ln n_2})(n_1^{-1/2} + n_2^{-1/2})\Big).$$

Step 4: Tail evaluation.

Using

$$\phi(\sqrt{2\ln n_2}) = \frac{1}{\sqrt{2\pi}} e^{-\ln n_2} = O\left(\frac{1}{n_2}\right),$$

we obtain

$$F_{\mathbf{W}}(\mathbf{w}(\hat{\theta})) - F_{\mathbf{W}}(\mathbf{w}(\theta)) = O_P\left(\frac{1}{n_2}(n_1^{-1/2} + n_2^{-1/2})\right).$$

Multiplying by n_2,

$$n_2\Big(F_{\mathbf{W}}(\mathbf{w}(\hat{\theta})) - F_{\mathbf{W}}(\mathbf{w}(\theta))\Big) = O_P(n_1^{-1/2} + n_2^{-1/2}) \;\longrightarrow\; 0.$$

Step 5: Conclusion.

Combining Steps 1 and 4 yields

$$P(\hat{\theta}) - P(\theta) \xrightarrow{P} 0,$$

as claimed. □